MTH 310 **Exam 2** Name:____________________

Spring, 2026 Sections 14.4–15.2

Instructions. (100 points) Sections 14.4–15.2. You may use a calculator. Show all work for credit.

(12pts) **1.** The two legs of a right triangle are measured as 5 cm and 12 cm with a possible error of at most 0.1 cm in each. Use differentials to estimate the maximum error in the calculated hypotenuse.

(12pts) **2.** Let $z = x^2y + xy^2$, where $x = s + 2t$ and $y = 2s - t$. Use the chain rule to find $\dfrac{\partial z}{\partial t}$ when $s = 1, t = 0$.

24 pts

(12pts) **3.** Consider $f(x, y) = x^2 e^{-y}$ at the point $P = (-2, 0)$.

(a) *(8 pts)* Find the rate of change of f at P in the direction toward the point $Q = (2, -3)$.

(b) *(4 pts)* What is the maximum rate of change of f at P, and in what **unit** direction does it occur?

(12pts) **4.** Find and classify all critical points of $f(x, y) = x^3 + y^3 - 3xy$.

24 pts

(14pts) **5.** Use Lagrange multipliers to find the maximum and minimum values of $f(x, y) = x^2y$ subject to the constraint $x^2 + y^2 = 3$.

(12pts) **6.** Set up, *but do not evaluate*, a single iterated integral for $\iint_D (x + y)\, dA$, where D is the region bounded by $y = x^2$ and $y = 2x$. Full credit requires correct limits of integration.

26 pts

(12pts) **7.** Evaluate $\int_0^8 \int_{\sqrt[3]{y}}^2 e^{x^4}\,dx\,dy$ by reversing the order of integration.

(14pts) **8.** Let $f(x,y) = x^2+y^2-2x-4y+1$ and let D be the closed triangular region with vertices $(0,0)$, $(4,0)$, and $(0,4)$. The only critical point of f is $(1,2)$, which lies inside D, and $f(1,2) = -4$.

Find the absolute maximum and minimum values of f on D. You must check all seven candidate locations.

26 pts